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SUPERSEDING
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MILITARY SPECIFICATION

ICE MAKING MACHINES, FLAKE, ELECTRIC, SELF-CONTAINED, 1000-POUND/24 HOURS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers automatic, electric motor driven, self-contained, mechanically refrigerated ice making machines with ice storage box or bin.

1.2 Classification. Ice making machines shall be of the following duties, classes, styles, and grades, as specified (see 6.2).

Duty A - Field (with shipping storage box)
  Class 1 - Air cooled condenser
    Style (a) - With cooling tower
    Style (b) - Without cooling tower

Duty B - Commercial (with ice storage bin)
  Class 1 - Air cooled condenser
  Class 2 - Water cooled condenser

Grade A - Corrosion-resisting steel
Grade B - Low carbon steel
Grade C - Aluminum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end ofthis document or by letter.

FSC 4110

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2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-F-1421	-	Fluorocarbon Refrigerants
HH-I-524	-	Insulation Board, Thermal (Polystryrene)
TT-E-529	-	Enamel, Alkyd, Semi-Gloss
WW-T-799	-	Tubing, Copper, Seamless, Water (for use with Solder, Flared, or Compression-Type Fittings)
PPP-P-600	-	Porcelain Enamel Products and Household Appliances, Electrical and Mechanical-Requirements for Packaging

MILITARY

MIL-P-514	-	Plates; Identification, Instruction, and Marking, Blank
MIL-T-704	-	Treatment and Painting of Material
MIL-V-173	-	Varnish, Moisture and Fungus Resistant, for the Treatment of Communications, Electronic, and Associated Electrical Equipment
MIL-F-3541	-	Fittings, Lubrication
MIL-B-11040	-	Belt, V; Engine Accessory
MIL-R-12323	-	Refrigerators and Related Equipment, Packaging and Packing of
MIL-C-24111	-	Chromium-nickel Alloy Steel Bars, Rod and Forgings, Corrosion Resistant (Precipitation Hardening)

STANDARDS

FEDERAL

FED-STD-H28	-	Screw Thread Standard for Federal Services
FED-STD-595	-	Colors

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference

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MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of
MS 17291	-	Receivers, Liquid, Refrigerant-12
MS 17797	-	Valves, Regulating, Fluid Pressure, Water, Refrigerant-12
MS 35845	-	Dehydrators, Desiccant, Refrigerant-12

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SOCIETY OF AUTOMOTIVE ENGINEERS INC.

AMS 5604 - Steel, Sheet, Strip and Place Corrosion Resistant

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096)

UNDERWRITERS' LABORATORIES, INC.

Standard No. 563 - Ice Makers 1970

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

RUBBER MANUFACTURER'S ASSOCIATION
Engineering Standards for Multiple V-Belt Drives

(Application for copies should be addressed to the Rubber Manufacturer's Association, 444 Madison Avenue, New York, NY 10022.)

AMERICAN SOCIETY FOR HEATING, REFRIGERATING AND AIR CONDITIONING ENGINEERS (ASHRAE)

Standard No. 29 - Methods of Rating and Testing Ice Makers - June 1956

(Application for copies should be addressed to the American Society of Heating, Refrigerating and Air Conditioning Engineers, 345 East 47th Street, New York, NY 10017.)

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AIR-CONDITIONING AND REFRIGERATION INSTITUTE (ARI)

ARI 710 - Standard for Liquid-Line Driers

(Application for copies should be addressed to the Air Conditioning and Refrigeration Institute, 1815 North Fort Myer Drive, Arlington, VA 22209).

NATIONAL SANITATION FOUNDATION (NSF)

Listing of Food Service Equipment
Standard No. 12 - Automatic Ice Making Equipment - June 1964

(Application for copies should be addressed to National Sanitation Foundation, P.O. Box 1468, Ann Arbor, MI 48106.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167	-	Corrosion-Resisting Chromium Steel Plate, Sheet and Strip
A 366	-	Steel, Carbon, Cold-Rolled, Commercial Quality
B 209	-	Aluminum-Alloy Sheet and Plate
D 3951	-	Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 70 National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02169.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

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3.2 Design. The ice making machine shall be a complete, self-contained unit, consisting of a condensing unit, an evaporator to which water is applied and frozen to form ice, means for harvesting and storing ice, a water circulating system, motor or motor drives, and all necessary piping, wiring, control and safety devices, and accessories required to provide an automatic operating ice maker with automatic control of ice level in storage bin or box (see 4.5.4). All parts capable of causing vibration shall be mounted on vibration elimination units to prevent the transmission of vibration during normal use. All parts of duty A machines shall be mounted so that continued vibration from transportation over rough terrain will not affect operation (see 4.5.2). All accessories, including fans, reduction gears, harvesting equipment, and pumps shall be direct drive design, V-belt driven, or chain driven from a reduction gear with protective guard. The ice making machine shall not leak refrigerant when tested as specified in 4.5.7.

3.2.1 Controls. A control shall automatically stop the ice making process when the predetermined level of ice in the ice storage box (duty A) or bin (Duty B) is reached, and automatically restart the ice making process when the amount of ice is reduced below the predetermined level. A water flow safety switch shall be provided that automatically stops operation of the ice flake machine in the event of an insufficient water supply and automatically resets with an adequate water supply.

3.2.2 Capacity. The ice making machine shall have a capacity to make and harvest ice weighing 22 to 30 pounds per cubic foot at a rate of not less than 1000 pounds in a 24-hour period, under table I conditions, when tested in accordance with 4.5.3.

TABLE I.- Capacity conditions

Duty	Class	Relative humidity (percent)	Ambient air temperature	Potable water supply temperature	Condenser inlet water temperature
A	1	40	100°F	90°F	Not applicable
B	1	40	100°F	90°F	Not applicable
B	2	40	90°F	90°F	90°F

3.2.3 Low temperature performance. When supplied with water at a temperature of 55°F, and operated at an ambient dry-bulb temperature of 55°F for a minimum of 2 hours the ice making machine shall produce ice of the quality specified in 3.2.2. During the harvest cycle the ice shall not adhere to or freeze to the evaporator or ice making section (see 4.5.8).

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3.2.4 Performance at maximum operating conditions. The ice making machine shall start and operate with maximum operating conditions at 90 percent and 110 percent of the rated name plate voltage (see 3.6.1.1.2) in an ambient temperature of 110°F for a minimum of 4 hours with the potable water temperature and condenser water temperature of 110°F (see 4.5.9).

3.3 Standard product. Ice making machines and storage bins delivered under this document for Duty B, and the equipment and component parts of ice machines for Duty A, shall be the manufacturer's standard commercial product, except for any changes necessary to comply with the specification requirements. Component parts of the ice making machines need not be the product of the same manufacturer.

3.4 Safety and sanitary standards. The ice making machines shall conform to the applicable requirements of UL Standard No. 563 for safety and NSF Standard No. 12 for sanitation.

3.4.1 UL certification. Prior to approval of the first article sample, if one is submitted, or prior to approval of the first shipment, the contractor shall submit to the contracting officer or his authorized representative, satisfactory evidence that the ice making machine he proposes to supply under this document meets the applicable requirements of UL Standard No. 563. Acceptable evidence of meeting the applicable requirements of this standard shall be the UL Listing Mark or a certified test report from a recognized independent testing laboratory, indicating that the ice making machine offered conforms to the applicable requirements of UL Standard No. 563.

3.4.2 NSF certification. The contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the ice making machine he proposes to furnish under this document meets the applicable requirements of NSF Standard No. 12 as follows:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished ice making machine shall be submitted prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, or

(2) A certification for the ice making machines issued by NSF under their special one-time contract evaluation/certification service shall be submitted prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, or

(3) A test report, acceptable to the Contracting Officer with the advice of the Army Surgeon General from an independent testing laboratory, indicating that the ice making machine has been tested and conforms to NSF.

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3.5 Materials (see 6.4) and components. Materials and components not definitely specified shall be of the quality normally used by the manufacturer for ice making machines provided the completed end items comply with all provisions of this document. Unless otherwise specified herein, commercial tolerances shall apply to metal thicknesses and diameters.

3.5.1 Refrigerants. The refrigerants shall conform to type 12 or 22 of BB-F-1421 for Duty A and type 12, 22, or 502 of BB-F-1421 for Duty B machines.

3.5.2 Copper tube. Hard copper tube shall be seamless conforming to type K or L of WW-T-799. Soft copper tube shall be seamless, standard commercial refrigeration tube.

3.5.3 Steel.

3.5.3.1 Corrosion-resisting. Corrosion-resisting steel shall conform to any AISI number in the 300 series of ASTM A167 or AMS 5604 or MIL-C-24111.

3.5.3.2 Carbon. Low carbon steel shall conform to ASTM A366.

3.5.4 Aluminum. Aluminum plate and sheet shall conform to alloy 3003-H14 or 3003-H24, bright finish of ASTM B209.

3.5.5 Insulation. Insulation shall be nonmoisture absorbing material; shall be homogeneous throughout; and shall have "K" factors specified below using 75°F mean temperature and expressed in terms of British thermal units (B.t.u.) per hour per square foot per inch of thickness per °F temperature difference. Loose fill type insulation will not be acceptable.

3.5.5.1 Pre-expanded insulation. Pre-expanded insulation shall be polystyrene material conforming to class 2, type I or II of HH-I-524.

3.5.5.2 Foamed-in-place insulation. Foamed-in-place insulation shall be polyurethane material having a manufacturer's commercial rated "K" factor of not more than 0.15, and a core density of not less than 1.0 pounds per cubic foot when tested as specified in 4.5.5.

3.5.5.3 Semi-rigid. Semi-rigid insulation shall be "B" fiber or finer industrial type fiberglass with a manufacturer's commercial rated "K" factor of not more than 0.250, with a manufacturer's rated density of at least 1.0 but not more than 3.0 pounds per cubic foot, and shall be held together with a thermosetting binder.

3.5.6 Fasteners. Screws, bolts, nuts, and other fasteners shall be corrosion-resisting steel or zinc, or nickel plated low carbon steel. Threads shall conform to FED-STD-H28.

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3.6 Construction. The ice making machine components, except storage box or bin, shall be totally inclosed in a structural frame and sheet metal housing with removable panels. The ice making machine shall show no signs of failure after a 3-inch drop when tested in accordance with 4.5.1. Joints between dissimilar metals which may become wet in service shall be protected against electrolytic or galvanic corrosion by proper selection of materials, coatings, isolation, gasketing, or similar means.

3.6.1 Condensing unit.

3.6.1.1 Compressor. The compressor for Duty A ice making machines shall be open type or semihermetic. The compressor for Duty B ice making machine shall be open type, hermetic, or semihermetic. Compressors shall have at least two cylinders, and shall be furnished with suction and discharge shut-off valves. The assembled refrigeration system shall be completely dehydrated and charged with the amount of refrigerant and oil necessary for operation.

3.6.1.1.1 Open type compressors. Open type compressors for Duty A ice flake machines shall be V-belt driven. V-belts shall conform to the Engineering Standards for Multiple V-belt Drives of the Rubber Manufacturer's Association. In addition, the belt or belts shall conform to MIL-B-11040. The seal of open type compressors shall be replaceable without the use of special tools and without regrinding of shaft shoulder. Compressors shall be equipped with means for determining the oil level in the crankcase.

3.6.1.1.2 Compressor motor. The compressor motor shall have sufficient capacity to start and operate the refrigerating system during the conditions specified in 3.2.4. Unless otherwise specified (see 6.2), motors shall be designed to operate on a nominal 208-volt, 60-hertz (Hz), 3-phase electrical power system. When frequency other than 60-Hz is specified, the motor shall be designed for the frequency cited. The motor mount on Duty A ice making machines shall include provisions for belt adjustment and locking in position using only standard handtools. The motors shall be equipped with a thermal overload protector on each phase.

3.6.1.2 Condenser. The Class 1 ice making machine condenser shall be air cooled, constructed of copper, aluminum, steel, or a combination thereof. The Class 2 ice making machine condenser shall be water cooled of the shell and coil type or the multipass shell and tube type with provisions for cleaning the water circuit without opening the refrigerant circuit. Shell shall be steel and tubes and coils shall be copper. The water cooled condenser shall be equipped with a water regulating valve as specified in MS 17797 or when refrigerant type 22 is used, an equivalent pressure-actuated automatic valve.

3.6.1.3 Wiring. Wiring design and practices shall be in accordance iwth NFPA 70.

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3.6.1.4 Safety device. A safety manually reset device shall be provided which will shut down the ice making machine when a loss of refrigerant results.

3.6.1.2.1 Receivers. Receivers for type 12 refrigerant, when used, shall be as specified in MS 17291. Receivers for type 22 refrigerant, when used, shall be fabricated from steel and shall have a 25-percent capacity greater than the volume of saturated liquid at the temperature of the refrigerant as it leaves the compressor. Receivers shall be fitted with a back seating outlet valve, a plugged 1/8 inch National Pipe Thread (NPT) opening and with a fusible plug.

3.6.2 Evaporator. Parts of the evaporator contacting ice shall be corrosion-resisting steel, anodized aluminum, or other nontoxic corrosion resistant material.

3.6.3 Cutting and harvesting blades. Cutting and harvesting blades and components shall be fabricated of material specified in 3.5.3.1.

3.6.4 Water circulator or pump. The water circulator or pump, when used, shall have a capacity adequate to provide continuously an equal volume of water over the entire ice making surface. Pumps shall be provided with corrosion-resistant drain plug or be self-draining.

3.6.5 Water tank. Water tanks shall be of material specified in 3 5.3.1 for Duty A and for Duty B shall be material in 3.5.3.1 or other suitable corrosion-resistant material.

3.6.6 Dehydrator. A dehydrator for refrigerant type 12 as specified in MS 35845 shall be installed in each open or semihermetic condensing unit. For refrigerant type 22, a dehydrator having a flare connection and conforming to ARI 710 shall be installed in each open or semihermetic condensing unit.

3.6.7 Tubing. Tube shall be as specified in 3.5.2. All tube connections shall be solder, except flared connections shall be provided on expansion valve, inlet and outlet connections, receiver outlet valve, and other locations normal to the manufacturer's standard product.

3.6 8 Cabinet. The cabinet for the Grade A machine shall be fabricated of the material specified in 3.5.3.1 having a nominal thickness of 0.0312 inch. The Grade B machine shall be fabricated from the material specified in 3.5.3.2 having a nominal thickness of 0.0299 inch. The cabinet for the Grade C machine shall be fabricated from the material specified in 3.5.4 having a nominal thickness of 0.040 inch. The cabinet shall have removable side panels. The frame shall be not less than 3/32 inch angle or channel steel, welded or bolted into an integral unit. The cabinet for the Duty B machines shall be mounted on skids or legs, a minimum of 4 inches high designed for mounting on or sealing directly to the floor, or designed for mounting on a suitable storage bin. Duty A machines shall be mounted on skids unless designed to be set on storage box (see 3.6.10.1).

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3.6.9 Cooling tower. When specified (see 6.2), a cooling tower shall be furnished for Duty B ice making machines. The tower shall have a rated B.t.u./hr. capacity of not less than the output capacity of the condensing unit of the ice making machine at the ambient temperatures specified.

3.6.10 Storage of ice.

3.6.10.1 Duty A. Each ice making machine for Duty A shall be furnished with a shipping container which shall also serve as an ice storage box during actual operation. The shipping container shall be provided with 6-inch high steel skids fastened to the bottom. The bottom is defined as the underside of the shipping container when in the proper operating position. Horizontal ice making machines (height less than length or width) shall be designed to be mounted on the ice storage box. Vertical ice making machines (height greater than length or width) shall be designed to be installed beside the ice storage box or mounted on the ice storage box. The operating height of the ice making machines with the storage box shall not exceed 75 inches. Since the ice making machine is to be shipped in the ice storage box, the ice storage box interior shall have dimensions approximately those of the outside of the ice making machine cabinet. The ice storage box shall be constructed as a standard refrigerator type ice storage chest and shall be fitted with a drainage opening and plug of nontoxic materials. The floor of the storage box shall be beveled to the drain. The outer shell shall be constructed of 0.0478-inch thick sheet steel and shall be sealed by welding. The ice storage box may be frame or frameless construction. Framing, when used, shall be steel. The inner shell shall be constructed of corrosion resisting steel specified in 3.5.3.1 with a nominal thickness of 0.0312-inch, sealed water-tight by soldering (see 4.4.2.3). The ice storage box, when constructed frameless, shall be reinforced on the outside with 0.185-inch steel angle or channel welded into an intergral unit. The ice storage box shall be provided with a gasketed ice chute located in or near the top with provisions for preventing ice collecting in the chute. The ice storage box shall have either a vertical inside sliding door or a flush swinging door located on the front or top to provide for ice removal and cleaning of the ice storage box. Each ice storage box shall be provided with a gasketed top equal in width and length to that of the ice storage box. The top of the ice storage box shall be provided with recessed handles.

3.6.10.2 Duty B.- Each Duty B ice making machine shall be provided with an ice storage bin. Unless otherwise specified (see 6.2), the ice storage bin shall have sufficient capacity for 12 hours of ice production. The ice storage bin shall be lined with corrosion-resisting steel specified in 3.5.3.1 or polyethylene plastic. The thickness of the liner shall be not less than 0.025 inch, except when foamed-in-place insulation is used the thickness of the liner shall be 0.0125 inch. A suitable insulated ice-removal-door shall be provided and equipped with the necessary handle, latch, and trim. The ice storage bin shall have a drainage opening fitted with a welded-in corrosion-resisting steel or brass fitting with female pipe thread. The inner liner shall not leak when tested as specified in 4.4.2.3.

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3.6.10.3 Insulation. The ice storage bin for the ice making machines shall be insulated with material specified in 3.5.5. The insulation thickness shall be determined by the manufacturer so that the computed overall heat transfer coefficient, "U", (B.t.u./hr. per square foot, per ^{o}F) is not more than 0.10 for the top and four sides, and not more than 0.07 for the bottom of the ice storage box. The computed "U" value shall be the "K" of the insulation value divided by the thickness of the insulation used. Rigid insulation, preformed or foamed-in-place, shall be used for the bottom of the ice storage box and may be used throughout the ice storage box. When a semirigid insulation is used for the top and four sides, it shall be cut slightly larger and thicker than the shape and space to be filled so as to require forcing into place.

3.6.11 Lifting attachments. Each duty A ice making machine and ice storage box shall be provided with attachments to enable the ice making machine contained therein to be lifted. Each individual attachment on the ice storage box shall be of sufficient strength to support the total weight of the ice storage box and the inclosed ice making machine. The lifting attachments shall be constructed to withstand handling conditions which may be encountered, including rapid lowering and braking and the unit shall operate and have no leaks when tested as specified in 4.5.6.

3.6.12 Fungus resistance. When specified (see 6.2), electrical connections, including terminal and circuit connections, components, and circuit elements shall be coated with varnish conforming to MIL-V-173 except that:

(1) Components and elements inherently inert to fungi or hermetically sealed need not be treated.

(2) Components and elements whose operation will be adversely affected by the application of varnish shall not be treated.

(3) One percent of copper-8-quinolinolate (by weight based on the nonvolatile content of the varnish) may, at the contractor's option, be substituted for the salicylanilide specified in MIL-V-173, provided that the varnish conforms to MIL-V-173 in all other respects.

3.6.13 Lubrication. All moving parts and other surfaces requiring lubrication shall be provided with accessible means for applying the required lubricants. Fan shaft bearings shall be lubricated from one general location. The equipment shall be designed to operate efficiently and satisfactorily when lubricated with standard Military lubricants (see 6.6). Grease fittings shall conform to MIL-F-3541. Bearings requiring lubrication shall be of such design and construction that lubrication cannot leak, drip, or be forced into contact with the ice.

3.6.14 Caution plates. Where the use of high-pressure lubricating equipment (1,000 pounds per square inch gage or higher) will damage grease seals or other parts of duty A machines, suitable warning shall be affixed to the equipment in a conspicuous location.

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3.6.15 Initial lubrication service. Unless otherwise specified (see 6.2), equipment shall be serviced, before delivery to the Government, with winter grades of standard Army or Navy lubricants designated for use in the temperature range of $0^{o}F$ to plus $32^{o}F$. After the lubrication services, each item of equipment shall be tagged in a conspicuous place to indicate the temperature range and grades of lubricant used. Points requiring lubrication shall be accessible.

3.7 Finish. The Duty A ice making machine and shipping-storage box shall be treated and painted in accordance with type A of MIL-T-704. The enamel shall conform to TT-E-529. Unless otherwise specified (see 3.7.1 and 6.2), the color of Duty A ice making machines shall be olive drab conforming to number 24087 of FED-STD-595. Burrs, rough edges, and slivers shall be removed. The Duty B ice making machine and bin shall have matching finish and trim.

3.7.1 Marine Corps procurement. The ice making machines and shipping storage box shall be painted and treated in accordance with table II, type A of MIL-T-704, color to match Marine Corps Green No. 23. Color chips may be obtained from the Commandant of the Marine Corps (see 6.8).

3.8 Electromagnetic compatibility. When specified (see 6.2), the ice making machines shall be designed and equipped for electromagnetic compatibility in accordance with class C3, UMO5, Group 1 of MIL-STD-461 (see 4.5.6).

3.9 Instruction plate. Each ice making machine shall be equipped with an instruction plate suitably located and securely affixed, describing procedures for operating and servicing the ice making machine. The plate shall be 0.028-inch thick and shall be constructed of material that is compatible electrochemically and not subject to galvanized or corrosive action. The instruction plate shall include the following statement: "The ice storage box must be periodically cleaned and sanitized in accordance with instructions provided by local medical authorities."

3.10 Identification marking.

3.10.1 Duty A ice making machine. The identification marking for Duty A ice making machine shall be on a name plate conforming to MIL-P-514, conspicuously located and permanently attached. The following nomenclature, shall appear on the plate: Ice Making Machine, Flake, Electric, Self-Contained 1,000 lbs day and the class.

3.10.1.1 Marine Corps procurement. Marine Corps data plates for the ice making machine will be furnished upon request to the Commandant of the Marine Corps (see 6.8). Data applicable to spaces 1, 3, 8, 9, and 10 with instructions shall be furnished. The contractor shall be required to imprint data on the applicable spaces and permanently affix plates to the equipment in a conspicuous location.

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3.10.2 Duty B ice making machine. The identification marking for the Duty B ice making machine shall be on a name plate 0.028-inch thick minimum, conspicuously located and permanently affixed. The name plate shall be constructed of material that is compatible electrochemically and not subject to galvanic or corrosive action. The manufacturer's standard name plate data shall appear on the name plate.

3.10.3 Wiring diagram. A wiring diagram shall be permanently affixed in the machine compartment of the ice making machine so that it may be read from the front. The diagram shall be waterproof and shall portray each electrical component and its terminals by sketch, outline, or name, and not by electrical symbol alone, showing the wiring of the ice making machine with the wires coded for identification.

3.10.3.1 Identification marking for electrical wiring. Identification markings shall be permanently placed on the insulation of all wiring at not more than 2-inch intervals, over the last 6 inches of length, identical at both ends. The markings shall be color coded or number coded.

3.11 Workmanship. The finished ice making machines shall be clean with all forming and cutting done accurately. All scale or flux deposits shall be removed from the finished welds. All exposed welds shall be ground smooth to match the adjacent surfaces. All nuts, bolts, and screws shall be tight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required it shall be examined for the defects listed in table II and for dimensions specified, and tested as specified in 4.5.1, 4.5.2 (for duty A machines), 4.5.3, 4.5.4, 4.5.6, 4.5.8, 4.5.9, 4.5.10, and 4.5.11. Failure to pass any test specified shall be cause for rejection of the preproduction sample. The presence of any defect listed in table II or any dimensions not within specified requirements shall be cause for rejection of the preproduction unit.

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4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the require-ments of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 In-process inspection.

4.4.2.1 In-process examination. Examination shall be made of the following operations to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be made to the items affected and to the operation.

(a) Joints between dissimilar metals are as specified in 3.6.
(b) Insulation is installed in accordance with 3.6.10.3.
(c) Fungus resistance is in accordance with 3.6.12, when applicable.
(d) Welding is in accordance with 3.11.

4.4.2.2 Foamed-in-place insulation density testing. Prior to each day's production, one free foam sample of the insulation specified in 3.5.5.2, a minimum size 8-inch cube, shall be taken. From the center of the sample, a 6-inch cube shall be cut and weighed. This weight, 1/8 of a cubic foot, shall be multiplied by 8 to determine conformance to the density requirements of 3.5.5.2. In addition, a daily record of the foam insulation shall be kept as follows and made available to the contracting officer:

Calibration of mix (mix proportions)
Amount of flow (per box)
Time of flow (per box)

Inability of the insulation to meet the density requirements of 3.5.5.2 shall be cause for correction of the operation and retesting.

4.4.2.3 Inner liner leakage testing. Prior to assembly in the storage box or ice bin, the liner shall be tested for leakage by placing a plug in the drain of the liner, placing the liner in a jig, filling the liner with water, and then examining the liner for signs of leaks. Any failure to conform to the leakage requirements in 3.6.10.1 or 3.6.10.2 shall be classified as a defect. The lot size shall be expressed in units of liners. The sample unit shall be one liner. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5.

4.4.3 End item visual examination. The end item shall be examined for defects listed in table II. The lot size shall be expressed in units of ice making machines of one duty, class, and grade only. The sample unit shall be

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one ice making machine. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not color specified		X
	Burrs, rough edges, or slivers	X	
Design, construction, and workmanship (general)	Any component missing or not type specified	X	
	Any component fractured, split, punctured, torn, dented, creased, loose, bowed, or sprung	X	
Wiring	Wiring diagram not affixed as specified		X
	Wires not coded or marked as specified		X
	Lubrication fittings not painted red		X
Marking for identification	Missing, incomplete, not legible, not specified type, not affixed or marked in prescribed manner, or not of specified material		X

4.4.4 End item dimensional examination. The end item shall be examined for conformance to specified dimensions. Any dimension not as specified shall be classified as a defect. The lot size shall be expressed in units of ice making machines of one duty, class, and grade only. The sample unit shall be one ice making machine. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing.

4.4.5.1 Initial production item testing. When a first article is not required, the initial production item shall be tested as specified in 4.5.2 (applicable to Duty A only), 4.5.7, 4.5.8, and 4.5.9. Any test failure shall be cause for rejection of the lot.

4.4.5.2 Drop, capacity, and strength testing. The end item shall be drop tested as specified in 4.5.1, capacity tested as specified in 4.5.3, and lifting attachment strength (applicable to duty A only) tested as specified in

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4.5.5. The lot size shall be expressed in units of end items. The sample unit shall be one end item. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

4.4.5.3 Operational and refrigerant leakage testing. After completion of testing as specified in 4.4.5.1 and 4.4.5.2, every ice making machine in acceptable lots shall be operational tested as specified in 4.5.4 and refrigerant leakage tested as specified in 4.5.7. Any ice making machine failing either test shall be rejected.

4.4.6 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of MIL-R-12323.

4.4.7 Standard compliance. Proof of compliance with the requirements of UL , as applicable to 3.4.1, and NSF as applicable to 3.4.2, shall be submitted to the Government representative.

4.5 Methods of inspection.

4.5.1 Drop test. The ice making machine shall be raised once on each side and dropped onto a concrete floor from height of 6 inches and then operated to determine compliance with 3.6. Inability of the unit to operate through one ice making and harvesting cycle or presence of refrigerant leaks greater than determined in 4.5.7 shall constitute failure of this test.

4.5.2 Vibration and impact test. The complete unit, packed and packaged at levels A, B, or C as specified, shall be tested according to levels, A, B, or C, respectively, of PPP-P-600 for vibration and impact to determine compliance with vibration requirements of 3.2 for duty A machines. Inability of the unit to operate through one ice making and harvesting cycle or presence of refrigerant leaks greater than determined in 4.5.7 shall constitute failure of this test.

4.5.3 Capacity tests.- The ice making machine shall be tested in accordance with the methods prescribed in ASHRAE Standard No. 29, and conditions specified herein to determine compliance with 3.2.2. Inability of the unit to meet the capacity requirements shall constitute failure of this test.

4.5.4 Operational test.- The ice making machine shall be operated through one ice making and harvesting cycle to determine compliance with operational requirements of 3.2. Inability of the unit to operate through one ice making and harvesting cycle shall constitute failure of this test.

4.5.5 Strength test for lifting attachments.- The Duty A ice making cabinet shall be placed in the ice storage box and the entire unit lifted by one lifting attachment on the ice storage box. The test shall be repeated for each lifting attachment. All lifting attachments shall then be engaged and the ice storage box, with ice making cabinet inside, shall be raised 3 feet

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and lowered in a free fall and then braked suddenly to determine compliance with 3.6.11. The unit shall then be operated. Following this part of the test, the ice making cabinet shall be removed from the ice storage box and the tests repeated on the lifting rings of the ice making machine. Inability of the lifting attachments to hold the weight, unit to operate, or refrigerant leaks shall constitute failure of this test.

4.5.6 Electromagnetic compatibility test. When electromagnetic compatibility is required, the first article sample or initial unit of production, as applicable, shall be tested by the contractor in accordance with test methods of MIL-STD-462 for UMO5 requirements. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the Contracting Office written certification that the Interference Control Plan, the EMI/EMC Test Plan, the electromagnetic test report and the requirements meet MIL-STD-461.

4.5.7 Refrigerant leakage test. After tests in 4.5.1, 4.5.2, and 4.5.5, when performed, and after operational test in 4.5.4, the refrigeration system shall be tested with an electronic leak detector. The finding of leakage in any unit which is greater than 0.5 ounce per year shall constitute failure of this test.

4.5.8 Low temperature performance test. The ice making machine shall operate for 4 hours under the conditions specified in 3.2.3. Inability of the unit to produce the quality of ice specified in 3.2.2 or any freeze up in the evaporator or ice making section shall constitute failure of this test.

4.5.9 Maximum operating condition test. The ice making machine shall start and operate for 4 continuous hours under the operating conditions specified in 3.2.4. During the test measure and record the maximum amperes drawn by the compressor motor. Motor load in excess of its continuous service factor load in amperes or motor temperature rise in excess of rated temperature rise shall constitute failure of this test.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking for levels A, B, or C shall be in accordance with MIL-R-12323, (see 6.2). For Marine procurement (see 3.7.1).

6. NOTES

6.1 Intended use. Duty A ice making machines are intended for use by troops in the field. Duty B ice making machines are intended for use in military messes, hospitals, and laboratories.

6.2 Ordering data. Acquisition documents should specify the following:

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a. Title, number, and date of this document.
b. Duty, class, style, and grade desired (see 1.2).
c. When a first article is required (see 3.1, 4.2, and 6.3).
d. Type of refrigerant required (see 3.5.1).
e. Electrical characteristics of compressor motor, if other than specified (see 3.6.1.1.2).
f. Whether a cooling tower is required for duty B machines (see 3.6.9).
g. Capacity of duty B storage bin, if other than specified (see 3.6.10.2).
h. If fungus treatment is required (see 3.6.12).
i. If initial lubrication service is waived (see 3.6.15).
j. Color of duty A machines, if other than specified (see 3.7).
k. When electromagnetic compatibility is required (see 3.8 and 4.5.6).
l. Selection of applicable levels of preservation and packaging, and packing (see 5.1).
m. For special Marine painting and packaging consideration (see 3.7.1 and 5.1).
n. For duty A machines, whether shop drawings are required, and to whom sent (see 6.6).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The first article should consist of one unit. The contracting officer should include specific instructions regarding arrangements for inspection and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.5).

6.5 Standard lubricants. When requested by the contractor, a list of standard Military lubricants applicable to the equipment covered by this document will be furnished.

(a) For Army use, this information is contained in Department of the Army Supply Bulletin SB 38-5-3, "List of Standard Lubricants, Hydraulic Fluids, Liquid Fuels and Preservative Materials."

(b) For Navy use, this information is contained in the Department of Defense Section of the Federal Supply Catalog, C9100-IL Identification List.

6.6 Engineering design conference. The following information should be included in the invitation for bid. Within 10 days after award of contract and prior to any work on the preproduction model including ordering of materials, the contractor may request the Contracting Officer, in writing, to arrange an engineering design conference to discuss any questions concerning the details of Government drawings, instructions, and technical requirements

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of this document. This conference is to assist the contractor in interpretation of the document, and agreements made shall neither obligate the Government to accept equipment that is not in accordance with this document nor relieve the supplier of his obligation to supply the equipment specified.

6.7 Drawings. Shop drawings for duty A ice making machines should be prepared by the contractor and made available for use by the Government in the inspection and testing of the preproduction sample (see 3.1). The drawings should be in sufficient detail for necessary fabricating, assembling, and finishing. When specified (see 6.2), one complete set of these drawings together with a copy of instructions should be furnished by the contractor on the initial contract for any particular model machine to activity designated by the Contracting Officer.

6.8 Marine Corps procurement. Information regarding Marine Corps procurement may be obtained from the following: Commandant of Marine Corps (Code C56) Headquarters U.S. Marine Corps, Washington, D.C. 20380.

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Preparing activity:

Army - GL

Project No. 4110-0375

Review activities:

Army - MD, ME
Navy - MS
DLA - GS

User activity:

Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-163416	Ice Making Machines, Flake, Electric, Self-Contained, 1000-Pound/24 Hours

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 20 Nov 84

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